Problem Set 7: Gene Expression

Due: Sunday, November 29 at 11:59 PM **Value: 25 points** **Format: Five-question problem set**

Purpose

Use sequence information and experimental evidence to reason about transcription, RNA processing, translation, and regulation.

Your task

- Q1 (5): Transcribe the template DNA 3'-TAC GGA TTT CCA-5' and translate the resulting mRNA using a codon table.
- Q2 (5): Predict how a splice-site mutation could change mature mRNA and protein without altering promoter activity.
- Q3 (5): Compare a promoter mutation, nonsense mutation, and proteasome inhibitor by expected mRNA and protein abundance.
- Q4 (5): Explain why ribosome profiling and RNA sequencing answer different questions.
- Q5 (5): Interpret the expression table and identify the strongest evidence for translational regulation.

Provided evidence or data

Condition	mRNA abundance	Protein abundance
Control	1.0	1.0
Treatment A	3.8	3.6
Treatment B	1.1	4.2

Submit

- One PDF with sequence directionality and reasoning shown.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.